

TC_DIO_001

TC_DIO_001

Test cases

Defects

Team stats

Timeline

Search

Add filter

Select all

0 cases | 0 suites

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TC_DIO_001

TC_DIO_001

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Run Again

Mute

Execution

Run History

Retries

Case Description

Verify that the DIO module initializes correctly using the configuration array.

Case Pre-Conditions

Microship Studio and Proteus simulation are running and ready

Priority

High

Configure the DIO pins direction array

Input data

DIO_PinsStatusArr[TOTAL_PINS]

Call DIO_Init() in the main function to initialize the pin directions

Compile the code in Microship Studio

Run proteus simulation using the hex file generated

Open "Watch window" from 'debug' tab in proteus and add the following parameters in data section

Run attachments

```

const DIO_PinStatus_type DIO_PinsStatusArr[TOTAL_PINS]={
    INPUT, /* Port A Pin 0 ADC0*/
    OUTPUT, /* Port A Pin 1 ADC1*/
    OUTPUT, /* Port A Pin 2 */
    OUTPUT, /* Port A Pin 3 */
    OUTPUT, /* Port A Pin 4 */
    OUTPUT, /* Port A Pin 5 */
    OUTPUT, /* Port A Pin 6 */
    OUTPUT, /* Port A Pin 7 ADC7*/
    OUTPUT, /* Port B Pin 0 */
    OUTPUT, /* Port B Pin 1 */
    OUTPUT, /* Port B Pin 2 / INT2*/
    OUTPUT, /* Port B Pin 3 /OC0*/
    OUTPUT, /* Port B Pin 4 /ss*/
    OUTPUT, /* Port B Pin 5 //mosi*/
    OUTPUT, /* Port B Pin 6 /miso*/
    OUTPUT, /* Port B Pin 7 clk*/
    OUTPUT, /* Port C Pin 0 */
    OUTPUT, /* Port C Pin 1 */
    OUTPUT, /* Port C Pin 2 */
    OUTPUT, /* Port C Pin 3 */
    OUTPUT, /* Port C Pin 4 */
    OUTPUT, /* Port C Pin 5 */
    OUTPUT, /* Port C Pin 6 */
    OUTPUT, /* Port C Pin 7 */
    OUTPUT, /* Port D Pin 0 */
    OUTPUT, /* Port D Pin 1 */
    INPUT, /* Port D Pin 2 /INT0*/
    INPUT, /* Port D Pin 3 / INT1 */
    OUTPUT, /* Port D Pin 4 OC1B*/
    INPUT, /* Port D Pin 5 OC1A*/
    INPUT, /* Port D Pin 6 / ICP*/
    INPUT, /* Port D Pin 7 */
};

```

Watch Window

Name	Address	Value	Watch Expression
DDRA	0x003A	0xFE	
DDA0	0x001A	0	
DDA1	0x001A	1	
DDA2	0x001A	1	
DDA3	0x001A	1	
DDA4	0x001A	1	
DDA5	0x001A	1	
DDA6	0x001A	1	
DDA7	0x001A	1	
PORTA	0x003B	0x01	
PORTA0	0x001B	1	
PORTA1	0x001B	0	
PORTA2	0x001B	0	
PORTA3	0x001B	0	
PORTA4	0x001B	0	
PORTA5	0x001B	0	
PORTA6	0x001B	0	
PORTA7	0x001B	0	
DDRB	0x0037	0xFF	
DDB0	0x0017	1	
DDB1	0x0017	1	
DDB2	0x0017	1	
DDB3	0x0017	1	
DDB4	0x0017	1	
DDB5	0x0017	1	
DDB6	0x0017	1	
DDB7	0x0017	1	
PORTB	0x0038	0x00	
PORTB0	0x0018	0	
PORTB1	0x0018	0	
PORTB2	0x0018	0	
PORTB3	0x0018	0	
PORTB4	0x0018	0	
PORTB5	0x0018	0	
PORTB6	0x0018	0	
PORTB7	0x0018	0	
DDRC	0x0034	0xFF	
DDC0	0x0014	1	
DDC1	0x0014	1	
DDC2	0x0014	1	
DDC3	0x0014	1	
DDC4	0x0014	1	
DDC5	0x0014	1	
DDC6	0x0014	1	
DDC7	0x0014	1	
PORTC	0x0035	0x1C	
PORTC0	0x0015	0	
PORTC1	0x0015	0	
PORTC2	0x0015	1	
PORTC3	0x0015	1	
PORTC4	0x0015	1	
PORTC5	0x0015	0	
PORTC6	0x0015	0	
PORTC7	0x0015	0	
DDRD	0x0031	0x13	
DDD0	0x0011	1	
DDD1	0x0011	1	
DDD2	0x0011	0	
DDD3	0x0011	0	
DDD4	0x0011	1	
DDD5	0x0011	0	
DDD6	0x0011	0	
DDD7	0x0011	0	
PORTD	0x0032	0xEC	
PORTD0	0x0012	0	
PORTD1	0x0012	0	
PORTD2	0x0012	1	
PORTD3	0x0012	1	
PORTD4	0x0012	0	
PORTD5	0x0012	1	
PORTD6	0x0012	1	
PORTD7	0x0012	1	

| Page 1

TC_DIO_002

Case Description

Verify that **DIO_SetPin** sets the voltage level of a given pin.

Case Pre-Conditions

- DIO module initialized with all pins in **OUTPUT** mode.
- Proteus simulation running with the .hex file generated in TC_DIO_001

Priority

↑ High

- ✓ use for loop for all the pins to Call **DIO_SetPin(PIN, HIGH)** ✓  1

Attachments

 main.png

- ✓ Check 'Watch Window' in Proteus to watch pin values after being set to 'HIGH' ✓  2 

Input data

PINA (0-7)

PINB (0-7)

PINC (0-7)

PIND (0-7)

Expected result

All 32 PIN values are **HIGH**

Attachments

 A%26B_2.png

 C%26D_2.png

- ✓ use for loop for all the pins to Call **DIO_SetPin(PIN, LOW)**

- ✓ Check 'Watch Window' in Proteus to watch pin values after being set to 'LOW' ✓  2 

Input data

PINA (0-7)

PINB (0-7)

PINC (0-7)

PIND (0-7)

Expected result

All 32 PIN values are **LOW**

Attachments

 A%26B_3.png

 C%26D_3.png

Run attachments

```

////////////////////////////////////
int main()
{
    DIO_Init();
    for (int i=0; i<32; i++)
    {
        DIO_SetPin(i, HIGH);
    }

    while(1)
    {
        //
    }
}
////////////////////////////////////

```

TC_DIO_002 main.png

Watch Window

Name	Address	Value	Wate...
☒ DDRA	0x003A	0xFF	
DAA0	0x001A	1	
DAA1	0x001A	1	
DAA2	0x001A	1	
DAA3	0x001A	1	
DAA4	0x001A	1	
DAA5	0x001A	1	
DAA6	0x001A	1	
DAA7	0x001A	1	
☒ PINA	0x0039	0xFF	
PINA0	0x0019	1	
PINA1	0x0019	1	
PINA2	0x0019	1	
PINA3	0x0019	1	
PINA4	0x0019	1	
PINA5	0x0019	1	
PINA6	0x0019	1	
PINA7	0x0019	1	
☒ DDRB	0x0037	0xFF	
DDB0	0x0017	1	
DDB1	0x0017	1	
DDB2	0x0017	1	
DDB3	0x0017	1	
DDB4	0x0017	1	
DDB5	0x0017	1	
DDB6	0x0017	1	
DDB7	0x0017	1	
☒ PINB	0x0036	0xFF	
PINB0	0x0016	1	
PINB1	0x0016	1	
PINB2	0x0016	1	
PINB3	0x0016	1	
PINB4	0x0016	1	
PINB5	0x0016	1	
PINB6	0x0016	1	
PINB7	0x0016	1	
☒ DDRC	0x0034	0xFF	

TC_DIO_002 HIGH DDR and HIGH PIN for PORT A & B

Watch Window

Name	Address	Value	Wate...
☒ DDRC	0x0034	0xFF	
DDC0	0x0014	1	
DDC1	0x0014	1	
DDC2	0x0014	1	
DDC3	0x0014	1	
DDC4	0x0014	1	
DDC5	0x0014	1	
DDC6	0x0014	1	
DDC7	0x0014	1	
☒ PINC	0x0033	0xFF	
PINC0	0x0013	1	
PINC1	0x0013	1	
PINC2	0x0013	1	
PINC3	0x0013	1	
PINC4	0x0013	1	
PINC5	0x0013	1	
PINC6	0x0013	1	
PINC7	0x0013	1	
☒ DDRD	0x0031	0xFF	
DDD0	0x0011	1	
DDD1	0x0011	1	
DDD2	0x0011	1	
DDD3	0x0011	1	
DDD4	0x0011	1	
DDD5	0x0011	1	
DDD6	0x0011	1	
DDD7	0x0011	1	
☒ PIND	0x0030	0xFF	
PIND0	0x0010	1	
PIND1	0x0010	1	
PIND2	0x0010	1	
PIND3	0x0010	1	
PIND4	0x0010	1	
PIND5	0x0010	1	
PIND6	0x0010	1	
PIND7	0x0010	1	

TC_DIO_002 HIGH DDR and HIGH PIN for PORT C & D

Watch Window

Name	Address	Value	Wate...
DAA0	0x001A	1	
DAA1	0x001A	1	
DAA2	0x001A	1	
DAA3	0x001A	1	
DAA4	0x001A	1	
DAA5	0x001A	1	
DAA6	0x001A	1	
DAA7	0x001A	1	
☒ PINA	0x0039	0x00	
PINA0	0x0019	0	
PINA1	0x0019	0	
PINA2	0x0019	0	
PINA3	0x0019	0	
PINA4	0x0019	0	
PINA5	0x0019	0	
PINA6	0x0019	0	
PINA7	0x0019	0	
☒ DDRB	0x0037	0xFF	
DDB0	0x0017	1	
DDB1	0x0017	1	
DDB2	0x0017	1	
DDB3	0x0017	1	
DDB4	0x0017	1	
DDB5	0x0017	1	
DDB6	0x0017	1	
DDB7	0x0017	1	
☒ PINB	0x0036	0x00	
PINB0	0x0016	0	
PINB1	0x0016	0	
PINB2	0x0016	0	
PINB3	0x0016	0	
PINB4	0x0016	0	
PINB5	0x0016	0	
PINB6	0x0016	0	
PINB7	0x0016	0	
☒ DDRC	0x0034	0xFF	
☒ PINC	0x0033	0x00	

TC_DIO_002 HIGH DDR and LOW PIN for PORT A & B

Watch Window

Name	Address	Value	Wate...
☒ DDRC	0x0034	0xFF	
DDC0	0x0014	1	
DDC1	0x0014	1	
DDC2	0x0014	1	
DDC3	0x0014	1	
DDC4	0x0014	1	
DDC5	0x0014	1	
DDC6	0x0014	1	
DDC7	0x0014	1	
☒ PINC	0x0033	0x00	
PINC0	0x0013	0	
PINC1	0x0013	0	
PINC2	0x0013	0	
PINC3	0x0013	0	
PINC4	0x0013	0	
PINC5	0x0013	0	
PINC6	0x0013	0	
PINC7	0x0013	0	
☒ DDRD	0x0031	0xFF	
DDD0	0x0011	1	
DDD1	0x0011	1	
DDD2	0x0011	1	
DDD3	0x0011	1	
DDD4	0x0011	1	
DDD5	0x0011	1	
DDD6	0x0011	1	
DDD7	0x0011	1	
☒ PIND	0x0030	0x00	
PIND0	0x0010	0	
PIND1	0x0010	0	
PIND2	0x0010	0	
PIND3	0x0010	0	
PIND4	0x0010	0	
PIND5	0x0010	0	
PIND6	0x0010	0	
PIND7	0x0010	0	

TC_DIO_002 HIGH DDR and LOW PIN for PORT C & D

TC_DIO_003

[Execution](#) [Run History](#) [Retries](#)

Case Description

Verify that DIO_ReadPin returns the correct voltage level of a given pin.

Case Pre-Conditions

1. Open Microchip studio with the project loaded (or any IDE that has simulator debugger)
2. Run DIO_Init() while PINA1 is set as output
3. (If using optimizer) Turn off optimization in the project options

Priority

↑ High

- ✓ Initiate variable called **var** of type **DIO_PIN_VOLT**, which is an enumerated type representing the voltage level of a pin (either **LOW** or **HIGH**).
- ✓ Set PINA1 voltage to **LOW**
- ✓ **var** will receive the return from DIO_ReadPin() in two occasions, this is the first to read when PINA1 is set as **LOW**
- ✓ Set PINA1 voltage to **HIGH**
- ✓ this is the first to read using DIO_ReadPin when PINA1 is set as **HIGH**
- ✓ Run Simulator Debugger to check the value of DIO_ReadPin() after Step 3 and 5

2

Attachments

ReadPin_LOW.png

ReadPin_HIGH.png

Run attachments

The image displays two side-by-side screenshots of the Microchip Studio IDE, showing the execution of the TC_DIO_003 test case. Both screenshots show the same C code in the main function, with a breakpoint set at line 22. The code includes DIO_Init(), DIO_SetPin(PINA1, LOW), DIO_SetPin(PINA1, HIGH), and a while loop that reads the pin state into a variable 'var'.

Left Screenshot (TC_DIO_003 ReadPin_LOW): The Watch window shows the variable 'var' with a value of 'LOW'.

Name	Value	Type
var	LOW	DIO_PIN

Right Screenshot (TC_DIO_003 ReadPin_HIGH): The Watch window shows the variable 'var' with a value of 'HIGH'.

Name	Value	Type
var	HIGH	DIO_PIN

TC_DIO_003 ReadPin_LOW

TC_DIO_003 ReadPin_HIGH

TC_DIO_004

- Execution
- Run History
- Retries
- Defects

Case Description

Verify that `DIO_SetPort` sets the value of a given port. And Also verify that `DIO_ReadPort` returns the correct value of a given port.

Case Pre-Conditions

Run both Microchip Studio and Proteus simulation

Priority

Medium

Comment

Proteus simulation passed. However verifying `DIO_ReadPort` failed, as it returns enumerated value that contains only 4 values (0 to 3). It shall return unsigned char (u8).

Attachments

```
typedef enum{
    PA=0,
    PB,
    PC,
    PD
}DIO_Port_type;
```

```
}
DIO_ReadPort(
{
```

Run `DIO_Init()` but first make sure that we have a whole port as output (`PORTB`) in the `DIO_CFG.c`

Attachments

array.png

Set port B value to 73(0x49) then verify port value from both Proteus simulation and Microchip simulator debugger

Set port B with new value to 28(0x1C) then verify port value from both Proteus simulation and Microchip simulator debugger

Defect report

← Incorrect Return Type for DIO_ReadPort Function (TC_DIO_004)

Edit

Delete

Actual Result

The `DIO_ReadPort` function is returning a value of type `DIO_Port_type`, an enum with only 4 valid values. However, the function may return values beyond the enum's scope, which is not logical. The correct return type should be `unsigned char (u8)` to accurately reflect the expected range of values.

Attachments

```
typedef enum{
    PA=0,
    PB,
    PC,
    PD
}DIO_Port_type;
```

```
}
DIO_ReadPort(
{
```

Status

Open

Severity

Minor

Reporter

Ahmed Magdy

Assignee

Ahmed Magdy

Created

22 Jun 2024 05:05:06

```
typedef enum{
    PA=0,
    PB,
    PC,
    PD
}DIO_Port_type;
```

```
DIO_Port_type DIO_ReadPort(DIO_Port_type PT)
{
```

Watch Window

Name	Address	Value	Wate...
DDRA	0x003A	0xFE	
PINA	0x0039	0x01	
DDRB	0x0037	0xFF	
PINB	0x0036	0x49	
P... 0x0016	0x0016	1	
P... 0x0016	0x0016	0	
P... 0x0016	0x0016	0	
P... 0x0016	0x0016	1	
P... 0x0016	0x0016	0	
P... 0x0016	0x0016	0	
P... 0x0016	0x0016	1	
P... 0x0016	0x0016	0	
DDRC	0x0034	0xFF	
PINC	0x0033	0x00	
DDRD	0x0031	0x13	
PIND	0x0030	0xEC	